ICS 111 – Lab 4 Number Guessing Game

Upon completion of this assignment, you will be able to:

- Collect input from a user at runtime and save into variables.
- Generate a random number between a given range.
- Apply conditional statements to produce different outcomes based on a user's response.
- Apply looping statements to repeat code logic.

Program Objective:
Write a Java program that will play a number guessing game with a user. The computer will pick a random integer between 1 and 100. The user should be prompted to guess the number. After each guess, determine if the guess is correct, too high or too low or entered zero to exit. Display a relevant message for each scenario. The program (loop) ends when the user guesses the correct number or enters a zero (0) to end the game. When the game ends, display the computer's correct number.

Note: You may use the following expression to generate a random number between 1 and 100: `(int)(Math.random() * 100) + 1`

** Extra credit: Keep track of the number of tries and end the game if the user exceeds seven tries. Display number of tries at end of game.

Program Algorithm:

```
//import necessary classes



// Declare all variables and constants


// Create the Scanner "variable" object for input


// Display the program's purpose or introduction


// Generate and save a random integer between 1 - 100


// (LOOP) Continue to ask the user to enter a guess while their
answer is not correct and the number entered is not 0

     // collect user's guess
     // check if guess is 0, lower, higher or equal to computer #

// Display the computer's answer
```

Program Requirements:	**Points**
1) Imports necessary library for `Scanner` Class and creates a `Scanner` object.	0.5
2) Declares all variables and constants at top of program, using constants where appropriate.	0.5
3) Displays the program purpose/introduction.	1
4) Generate a random integer between 1 -100 and save value into a variable.	2
5) Use loop statements to continue to ask the user to enter a guess (integer) while their answer is not correct and the user guess is not equal to zero. a. Save user's guess to a variable b. Compare the user's guess to the computer's number. i. If the numbers match, display a congratulatory message ii. otherwise, display whether the guess is "too high" or "too low" and repeat the loop	6
6) When zero "0" is selected, program should exit and a relevant message should be displayed.	2
7) Display the correct answer.	2
8) Follows all programming standards as defined in the Programming Guidelines Version 02 document.	1
TOTAL POINTS	**15**

***Submit your completed** `.java` **file into Lamaku → Lab 4.**

See next page for an example →

Lab 4 Sample Output

```
----jGRASP exec: java NumberGuessingGame
Welcome to the Number Guessing Game!
Guess the number generated by the computer to win!

Enter an integer between 1 - 100 or 0 to quit: 50

Your guess is too low. Try again!
Enter an integer between 1 - 100 or 0 to quit: 75

Your guess is too high. Try again!
Enter an integer between 1 - 100 or 0 to quit: 60

Your guess is too low. Try again!
Enter an integer between 1 - 100 or 0 to quit: 67

Your guess is too low. Try again!
Enter an integer between 1 - 100 or 0 to quit: 73

Your guess is too high. Try again!
Enter an integer between 1 - 100 or 0 to quit: 70

Your guess is too high. Try again!
Enter an integer between 1 - 100 or 0 to quit: 68

Congratulations!!  You guessed it!
The computer's number was: 68
```

```
----jGRASP exec: java NumberGuessingGame
Welcome to the Number Guessing Game!
Guess the number generated by the computer to win!

Enter an integer between 1 - 100 or 0 to quit: 49

Your guess is too high. Try again!
Enter an integer between 1 - 100 or 0 to quit: 0

Well, that was difficult. Try again next time.
The computer's number was: 22

----jGRASP: operation complete.
```

Sample Output for Extra Credit:

```
----jGRASP exec: java NumberGuessingGameVerTwo
Welcome to the Number Guessing Game!
Guess the number generated by the computer to win!
You will have up to 7 tries to guess the correct number.

Enter an integer between 1 - 100 or 0 to quit: 50

Your guess is too low. Try again!
6 guesses left!

Enter an integer between 1 - 100 or 0 to quit: 75

Your guess is too low. Try again!
5 guesses left!

Enter an integer between 1 - 100 or 0 to quit: 76

Your guess is too low. Try again!
4 guesses left!

Enter an integer between 1 - 100 or 0 to quit: 77

Enter an integer between 1 - 100 or 0 to quit: 78

Your guess is too low. Try again!
2 guesses left!

Enter an integer between 1 - 100 or 0 to quit: 79

Your guess is too low. Try again!
1 guesses left!

Enter an integer between 1 - 100 or 0 to quit: 80

Your guess is too low. Try again!
0 guesses left!


You exceeded the maximum number of tries!
The computer's number was: 85
You made 7 guesses.
```

```
----jGRASP exec: java NumberGuessingGameVerTwo
Welcome to the Number Guessing Game!
Guess the number generated by the computer to win!
You will have up to 7 tries to guess the correct number.

Enter an integer between 1 - 100 or 0 to quit: 50

Your guess is too low. Try again!
6 guesses left!

Enter an integer between 1 - 100 or 0 to quit: 75

Your guess is too high. Try again!
5 guesses left
Enter an integer between 1 - 100 or 0 to quit: 65

Your guess is too high. Try again!
4 guesses left
Enter an integer between 1 - 100 or 0 to quit: 60

Your guess is too low. Try again!
3 guesses left!

Enter an integer between 1 - 100 or 0 to quit: 63

Your guess is too low. Try again!
2 guesses left!

Enter an integer between 1 - 100 or 0 to quit: 64

Congratulations!!  You guessed it!
The computer's number was: 64
You made 6 guesses.
```

```
----jGRASP exec: java NumberGuessingGameVerTwo
Welcome to the Number Guessing Game!
Guess the number generated by the computer to win!
You will have up to 7 tries to guess the correct number.

Enter an integer between 1 - 100 or 0 to quit: 50

Your guess is too high. Try again!
6 guesses left
Enter an integer between 1 - 100 or 0 to quit: 0

Well, that was difficult. Try again next time.
The computer's number was: 28
You made 2 guesses.
```